

STAT 4714 - Final Exam Study Guide

Question 1

A communication network has two parallel paths. The **upper path** has two routers in series with success probabilities 0.85 and 0.9. The **lower path** has three routers in series with success probabilities 0.8, 0.8, and 0.95. Routers fail independently. For the network to work, either the upper path must work entirely, the lower path must work entirely, or both.

1. What is the probability the entire network works?
 2. Given that the system works, what is the probability that the upper path has failed?
 3. Given that the system fails, what is the probability that the upper path also failed?
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Question 2

A sensor classifies electronic boards as “faulty.” 4% of boards are actually faulty. The sensor correctly identifies faulty boards 90% of the time, and it incorrectly labels a good board as faulty 7% of the time.

1. What is the probability a randomly selected board is labeled “faulty”?
 2. If a board is labeled “faulty,” what is the probability it is actually faulty?
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Question 3

The time between voltage spikes in a circuit is a random variable X with density

$$f(x) = 3(1 - x)^2, \quad 0 < x < 1.$$

1. What is the probability that $X > 0.6$?
 2. Given $X > 0.6$, what is the probability that $X < 0.8$?
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Question 4

Let X and Y represent the lifetimes of two system components with joint density

$$f(x, y) = 4e^{-2x-2y}, \quad x > 0, y > 0.$$

1. Compute $P(1 < X < 2 \mid Y = 1)$.
 2. Are X and Y independent?
 3. Compute σ_{XY} .
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Question 5

The lifetime T of a capacitor has CDF

$$F(t) = 1 - e^{-t/40}, \quad t > 0.$$

1. Determine the PDF.
 2. Compute $P(T > 90)$.
 3. Interpret $F(90)$ in plain language.
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Question 6

Random variables X and Y have joint density

$$f(x, y) = 6x, \quad 0 < x < y < 1.$$

Compute the correlation coefficient ρ_{XY} .

Question 7

A high-end chip has lifetime mean $\mu = 1000$ hours and standard deviation $\sigma = 80$ hours. Provide a **lower bound** for

$$P(820 < T < 1180)$$

and comment on the result.

Question 8

A robotics system uses 4 independent sensors, each with reliability 0.9. The system works if at least 2 sensors work.

1. Find the probability the system works.
 2. Is this sufficient for system reliability of 0.99? If not, how many such sensors would be needed with a majority-rule system?
 3. Suppose failures were positively correlated. How would this affect system reliability?
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Question 9

For each experimental design below, name the most appropriate discrete distribution and justify your choice in 1-2 sentences.

1. You flip a (possibly biased) coin n times; you record the number of heads.
 2. You flip a coin repeatedly until the first tail occurs; you record the number of flips.
 3. Network packets arrive randomly; you count how many arrive during a fixed 30-second interval.
 4. A device is tested until the 3rd failure occurs; you record the total number of tests.
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Question 10

For every use, a component fails independently with probability $p = 0.08$.

1. What is the probability of exactly 4 failures in 35 uses?
 2. What is the probability that the 4th failure occurs on the 35th use?
 3. Explain the conceptual difference between the two random variables modeled in parts (1) and (2).
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Question 11

A sensor misfires with probability $p = 0.03$ per transmission. We observe $n = 300$ transmissions.

1. Using a normal approximation, estimate $P(\text{at most 12 misfires})$. Show your continuity correction explicitly.
 2. State the conditions under which the normal approximation to the binomial is reasonable, and verify them here using n and p .
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Question 12

Briefly describe the conceptual relationship among the Poisson, exponential, and gamma distributions. Include:

1. How Poisson counts over an interval relate to exponential waiting times.
 2. How the gamma distribution arises in this setting and what its shape parameter represents.
 3. A concrete real-world example where all three appear together.
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Question 13

Let X be a uniform random variable over the interval $[0, 10]$. Suppose we learn that $X > 7$.

1. Compute $P(X > 9 \mid X > 7)$.
 2. Explain why the answer in (1) generally differs from $P(X > 9)$.
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Question 14

A cable of length 20 meters is inspected and 8 defects are found. Assume defects follow a Poisson process. For a new 5-meter cable, compute the probability of fewer than 2 defects.

Question 15

The lifetime (in years) of a certain type of cooling-fan module is exponential with average life $\beta = 3$ years. Assume modules fail independently. If 120 of these modules are installed in different systems, what is the probability that at most 30 fail during the first year?

Question 16

In a biomedical research study, it was determined that the survival time (in weeks) of an animal subjected to a certain exposure of gamma radiation has a gamma distribution with $\alpha = 5$ and $\beta = 10$.

1. What is the mean survival time of a randomly selected animal used in the experiment?
 2. What is the probability that an animal survives more than 30 weeks?
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Question 17

Consider the standard normal CDF and its quantile function.

1. Explain the difference between the CDF and quantile function in plain language.
 2. Give one practical task that uses the CDF and another that uses the quantile function.
 3. Suppose Z is a standard normal random variable. Write a formula for the 95% confidence interval of Z using the quantile function.
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Question 18

Runners pass a trail marker according to a Poisson process with rate $\lambda = 4$ per hour.

1. What distribution models the number of runners seen in a 45-minute window? State all parameter values.
 2. What distribution models the waiting time until the next runner? State all parameter values.
 3. Compute $P(\text{no runner within 30 minutes})$.
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Question 19

Briefly explain the following concepts/procedures:

1. Explain why the definition of a p -value depends on the alternative hypothesis. Illustrate with a generic test statistic and describe how the tail area is chosen.
 2. Explain why reporting a standard error (SE) conveys information that a confidence interval (CI) alone does not, and vice versa. Describe what each communicates about estimator variability and uncertainty.
 3. Outline the procedure you would follow to construct a bootstrap sampling distribution for an estimator $\hat{\theta}$ given only the observed data $\{x_1, \dots, x_n\}$. Be explicit about resampling, recomputing, and summarizing.
 4. Define Type I error, Type II error, and power. Then, discuss how changing the significance level α or the sample size n affects power, holding other parameters fixed.
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Question 20

Consider estimating the mean current draw of a device from n independent measurements with unknown variance. Describe, step-by-step, how you would construct an approximate sampling distribution for the sample mean using either (i) an assumption on the variance or (ii) a resampling technique. State any assumptions you need.

Question 21

A biomedical engineer measures the voltage drop across a certain implantable sensor during activation. Historically, under normal functioning conditions, voltage drop X is modeled as Normal with mean $\mu = 4.8$ volts and standard deviation $\sigma = 0.6$ volts. A new batch of sensors arrives from a different manufacturer. One sensor is

tested repeatedly and $n = 25$ activation-voltage measurements are collected, yielding a sample mean of $\bar{X} = 5.05$ volts. We want to test whether the new sensors differ from the historical functioning sensors.

We will conduct a two-sided hypothesis test:

$$H_0 : \mu = 4.8 \quad \text{vs.} \quad H_a : \mu \neq 4.8.$$

1. Describe, step-by-step, how you would approximate the p -value for this hypothesis test using parametric simulation. Your description should be specific and procedural such that another student could follow it exactly.
2. Explain the philosophical reason why it is important that the simulated samples are generated under the *null* model.
3. Suppose after simulation you compute that only 6.2% of simulated sample means are as or more extreme than the observed $\bar{X} = 5.05$. With 90% confidence, what is your decision?
4. Provide one advantage and one disadvantage of using simulation-based inference for hypothesis testing compared to using a closed-form theoretical test (e.g., Z test or t test).

Question 22

Consider the Simple Linear Regression (SLR) model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \sim \text{i.i.d. } N(0, \sigma^2).$$

1. Describe, in words, the differences between the quantities

$$\text{SST}, \quad \text{SSR}, \quad \text{SSE}.$$

Explain how the coefficient of determination R^2 relates to these quantities.

2. Describe the **bootstrap-based randomization test** for assessing whether a linear relationship exists between x and y . Explain the logic behind the test and outline the steps clear enough for a UVA grad to understand.
3. Outline the method of least squares estimation for the SLR model. What criterion is minimized, and what are the resulting formulas for $\hat{\beta}_0$ and $\hat{\beta}_1$?
4. Explain the difference between a **confidence interval** for the mean response and a **prediction interval** for a new individual response at the same value of x_0 . Which is wider, and why?
5. Describe how to obtain a confidence interval for the mean response at a new value x_0 . What quantities must be estimated?
6. Suppose you want to check whether the SLR model assumptions are violated. Describe how to use residual analysis to check for:
 - a. nonconstant variance (heteroscedasticity),
 - b. violations of normality.

Explain what patterns you would look for.

Question 23

An engineering team is analyzing the relationship between the continuous operating temperature in degrees Celsius (x) and the lifespan in hours (Y) of a new synthetic material used in aerospace components. They collect 50 independent data points at temperatures ranging strictly from 20°C to 80°C and fit a Simple Linear Regression (SLR) model. The resulting estimated regression line is $\hat{Y} = 1500 - 12x$ with a coefficient of determination $R^2 = 0.88$. Based on this high R^2 value, the lead engineer writes a report making the following two claims:

- **Claim 1:** “Because the R^2 is 0.88 (close to 1), this proves that a linear model is the correct structural choice for this relationship, and reviewing residual plots is unnecessary.”
- **Claim 2:** “We can safely use this regression equation to confidently predict the lifespan of the material when it is exposed to extreme atmospheric temperatures of -150°C .”

1. Identify the fundamental statistical flaw in Claim 1 and briefly explain why the reasoning is incorrect.
 2. Identify the fundamental statistical flaw in Claim 2 and briefly explain why the reasoning is incorrect.
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Question 24

A systems administrator wants to predict the server latency in milliseconds (Y) based on the active network traffic in gigabytes per second (x). To gather data, the administrator records the latency and traffic every minute for a continuous 48-hour period. They fit the following SLR model:

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

The regression output yields a statistically significant p -value for $\hat{\beta}_1$. However, a plot of the residuals versus the fitted values ($\hat{Y}$) reveals a distinct “funnel” or “fan” shape, where the vertical spread of the residuals becomes much wider as the fitted values increase. Then, the administrator uses the standard error from this SLR output to construct a 95% prediction interval for the latency during an upcoming high-traffic event.

1. Identify two major assumptions of the Simple Linear Regression model that are violated in this analysis.
2. Explain how these violations compromise the reliability of the administrator's 95% prediction interval.